

Sukanto Das

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Profile

Data Engineer with **6+ years of experience** building Python- and SQL-based data pipelines and scalable processing workflows across research and operational environments. Experienced in integrating heterogeneous datasets (EO, OSM, GTFS, NetCDF, WFS) and delivering analytics-ready data products through reproducible ETL processes. Strong background in workflow automation, data validation, and large-scale data integration, with domain expertise in geospatial and Earth observation data supporting climate and mobility analytics.

Core Skills

- **Data Engineering (Python & SQL):** ETL pipelines; workflow automation; reproducible data processing; data modeling; API ingestion; batch processing.
- **Databases & Storage:** PostgreSQL/PostGIS, DuckDB, GeoPackage, NetCDF; large-scale structured and spatial data handling.
- **Workflow Reliability & CI/CD:** Git, GitHub Actions; testing, validation, QA/QC, reproducible workflows, documentation.
- **Data Analytics & Feature Engineering:** clustering, anomaly detection, geo-temporal analytics, indicator generation.
- **Geospatial Data Processing:** GeoPandas, Rasterio, Shapely, GDAL; spatial ETL, accessibility analysis, large-volume batch processing.
- **Remote Sensing (Domain):** Sentinel-2, Landsat; LULC classification, change detection, preprocessing and feature extraction.

Professional Experience

Research Assistant — Leibniz Institute of Ecological Urban and Regional Development (IOER), Dresden, Germany *Jun 2023 – Nov 2025*

- Engineered **Python- and SQL-based ETL pipelines** (GeoPandas, GDAL, DuckDB) for processing large OSM, GTFS, WFS, and satellite datasets.
- Developed **unit-tested, reproducible workflows** integrating geospatial preprocessing, feature engineering, and ML/analytic pipelines.
- Conducted **geo-temporal mobility analysis** using clustering and anomaly detection; delivered map services and indicators supporting planning tasks.
- Performed workflow testing, validation, and quality checks to ensure reliable spatial outputs.
- Built interactive **Streamlit/Plotly dashboards** supporting operational monitoring and decision-making.
- Authored workflow documentation, metadata, and SOPs aligned with FAIR principles.

Data Management Officer — Wildlife Conservation Society, Dhaka, Bangladesh *Apr 2022 – Sep 2022*

- Maintained validated geodatabases and delivered **RS-derived LULC and change detection** products for conservation monitoring.

Senior Information Management Officer — iMMAP, Cox's Bazar, Bangladesh *May 2021 – Dec 2021*

- Produced **geospatial intelligence** integrating satellite imagery, surveys, and OSINT data for humanitarian operations.
- Streamlined **classification, feature extraction, and ETL workflows** to accelerate map production.
- Performed **feasibility and access analysis** for planning and coordination.

GIS Officer — Danish Refugee Council, Cox's Bazar, Bangladesh *Oct 2020 – May 2021*

- Automated geospatial workflows using **Python and ArcGIS Model Builder**, enabling rapid operational product delivery.
- Improved QA/QC processes for classification outputs and field data.

GIS/Remote Sensing Expert — TROYEE Associates, Dhaka, Bangladesh *Dec 2019 – Oct 2020*

- Performed **geospatial feasibility analysis** for infrastructure and land-use planning.
- Supervised GIS analysts; oversaw data processing, QC, and report/map preparation.

Research Associate — Bhumijo Limited, Dhaka, Bangladesh *Aug 2019 – Dec 2019*

- Conducted raster/vector analysis for spatial planning and public health; managed survey datasets.

Project Manager — Image Research & Consultancy Ltd., Dhaka, Bangladesh *Jan 2019 – Jul 2019*

- Led spatial research projects; supervised fieldwork and produced geospatial analytics for reporting.

Research Associate — REACH: IWFM-BUET, Dhaka, Bangladesh *Dec 2017 – Dec 2018*

- Conducted hydrological and geospatial analysis integrating lab, field, and satellite data.

Junior GIS/Remote Sensing Expert — Desh Upodesh Ltd., Dhaka, Bangladesh *Sep 2016 – Nov 2017*

- Processed multi-temporal RS/GIS datasets for feasibility and vulnerability assessments; produced analytical maps and reports.

Selected Projects

Urban Public Transit Frequency Indicator for Germany (Replication Dataset) — IOER Research Data Center *Data Manager, 2022–2025*

- Engineered the **end-to-end geospatial ETL workflow** for generating nationwide public transit frequency indicators using GTFS, OSM, and INSPIRE grid datasets.
- Processed **spatio-temporal GTFS feeds** (weekday/Sat/Sun + peak hours) and implemented weekly averaging, mode weighting, and QC routines.
- Applied **network-based interpolation** (QNEAT3, OSM routing) to compute 500 m walk-accessibility and derive 1 km grid frequency layers.
- Produced standardized **GeoPackage datasets, vector grids, and city-level statistics** with FAIR-compliant metadata for publication via IOER-FDZ.
- **Dataset DOI:** doi:10.71830/ABPCUS

Modeling Urban Fabrics for Net-Zero Intervention (M.Sc. Thesis) — IOER Dresden *May 2025 – Oct 2025*

- Developed **ETL pipelines** for POI and spatial datasets.
- Applied **Doc2Vec & LDA** for semantic classification of urban fabrics.
- Delivered reproducible workflows and interactive visualizations.

E-learning Tool for Spatial Change Analysis — IOER Dresden *Jan 2025 – Apr 2025*

- Integrated WFS, OSM, Sentinel-2, and GHSL datasets via OWSLib, Overpy, and GEE Python API; automated geoprocessing pipeline.
- Built web visualizations for settlement and mobility-change analysis.

Public Transit & Land-Use Dynamics Visualization — IOER Dresden *Jun 2023 – Dec 2024*

- Processed GTFS, OSM, and satellite imagery for accessibility and LULC dynamics.
- Designed interactive dashboards for researchers.

Climate Extremes Analysis Using CORDEX/ERA5-Land — TU Dresden *Apr 2023 – Sep 2023*

- Applied Python, NetCDF4, and CDO to calculate and visualize climate extreme indices.

Education

M.Sc. in Hydro Science & Engineering — TU Dresden, Germany *Oct 2022 – Nov 2025*
Thesis: *Modeling Urban Fabrics to Activate Net-Zero Intervention*

Bachelor in Urban & Regional Planning — KUET, Bangladesh *Feb 2012 – Jul 2016*
Thesis: *Investigating Coastal Livelihoods in Relation to Land-use/Land-cover Change Modeling*

Certifications

AI School: Scalable Data Analytics & HPC — ScaDS.AI Leipzig (2023)

AI School: Deep Learning, Data Visualization & LLMs — ScaDS.AI Leipzig (2025)

Citizenship: Bangladesh | **Languages:** English (Fluent); German (A2/B1)

References: Available upon request